

ASEN 5090 Lecture 16

Measurements Errors & Mitigation/Correction Methods

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RINEX Observation Codes

C = pseudorange (m)

L = carrier phase (cycles)

D = Doppler (Hz)

S = Signal Strength (Units vary)

Version 3.0 can handle newer measurements
and multi GNSS constellations

<https://files.igs.org/pub/data/format/rinex304.pdf>

GNSS System	Freq. Band /Frequency	Channel or Code	Observation Codes			
			Pseudo Range	Carrier Phase	Doppler	Signal Strength
GPS	L1/1575.42	C/A	C1C	L1C	D1C	S1C
		L1C (D)	C1S	L1S	D1S	S1S
		L1C (P)	C1L	L1L	D1L	S1L
		L1C (D+P)	C1X	L1X	D1X	S1X
		P (AS off)	C1P	L1P	D1P	S1P
		Z-tracking and similar (AS on)	C1W	L1W	D1W	S1W
		Y	C1Y	L1Y	D1Y	S1Y
		M	C1M	L1M	D1M	S1M
		codeless		L1N	D1N	S1N
	L2/1227.60	C/A	C2C	L2C	D2C	S2C
		L1 (C/A)+(P2-P1) (semi-codeless)	C2D	L2D	D2D	S2D
		L2C (M)	C2S	L2S	D2S	S2S
		L2C (L)	C2L	L2L	D2L	S2L
		L2C (M+L)	C2X	L2X	D2X	S2X
		P (AS off)	C2P	L2P	D2P	S2P
		Z-tracking and similar (AS on)	C2W	L2W	D2W	S2W
		Y	C2Y	L2Y	D2Y	S2Y
		M	C2M	L2M	D2M	S2M
		codeless		L2N	D2N	S2N
	L5/1176.45	I	C5I	L5I	D5I	S5I
		Q	C5Q	L5Q	D5Q	S5Q
		I+Q	C5X	L5X	D5X	S5X

Table 4 : RINEX Version 3.04 GPS Observation Codes

Pseudorange Model

$$\rho(t) = r(t, t - \tau) + c(\delta t_u(t) - \delta t^s(t - \tau)) + I_\rho(t) + T_\rho(t) + M_\rho(t) + \varepsilon_\rho(t)$$

- **GEOMETRIC RANGE**
- **CLOCKS**
- **RELATIVITY (NOT SHOWN IN EQUATIONS)**
- **IONOSPHERE (FREE ELECTRONS)**
- **TROPOSPHERE (NEUTRAL ATMOSPHERE)**
- **MULTIPATH**
- **MEASUREMENT NOISE**

GNSS Satellite Differential Code Bias (DCB)

%=BIA 1.00 CAS 22:266:60636 CAS 2022:263:00000 2022:264:00000 R 00005883

*-----

* Bias Solution INdependent EXchange Format (Bias-SINEX)

*-----

+FILE/REFERENCE

*INFO_TYPE_____

INFO_____

DESCRIPTION CAS, Chinese Academy of Sciences

INPUT CAS 1-day DCB solutions for G/R/C/E/J

OUTPUT CAS MGEX 1-day DCB solutions in Bias-SINEX format

CONTACT wangningbo@aoe.ac.cn or lizishen@aoe.ac.cn

SOFTWARE BDSMART-ION Version 1.10

HARDWARE Linux ion01 3.11.10-17-defaultx86_64

-FILE/REFERENCE

*-----

+FILE/COMMENT

- Daily multi-GNSS differential code biases (DCBs) are derived from observations of MGEX/IGMAS networks with local ionospheric TEC modeling technique entitled MGTS. A description of MGTS algorithms is provided in

Wang N., Yuan Y., Li Z., Montenbruck O., Tan B. (2016)

Determination of differential code biases with multi-GNSS observations, Journal of Geodesy, 90(3): 209-228.

- A zero-mean constellation condition is applied to separate satellite and receiver biases on a daily basis.

- Overview of the DCBs included in the file

GPS C1C-C1W, C2C-C2W, C2W-C2S, C2W-C2L, C2W-C2X

GPS C1C-C2W, C1C-C5Q, C1C-C5X, C1W-C2W

GLONASS C1C-C1P, C2C-C2P

GLONASS C1C-C2C, C1C-C2P, C1P-C2P

GALILEO C1C-C5Q, C1C-C6C, C1C-C7Q, C1C-C8Q, C1X-C5X, C1X-C7X, C1X-C8X

BDS-2 C2I-C7I, C2I-C6I

BDS-3 C1X-C5X, C1P-C5P, C1D-C5D

BDS-3 C1X-C6I, C1P-C6I, C1D-C6I, C2I-C6I

BDS-3 C1X-C7Z, C1X-C8X, C1X-C7D, C1P-C7D

QZSS C1C-C1X

QZSS C1C-C2L, C1C-C5X, C1C-C5Q, C1X-C2X, C1X-C5X

- Data archives

CAS: ftp://ftp.gipp.org.cn/product/dcb/

IGN: ftp://igs.ign.fr/pub/igs/products/mgex/dcb/

CDDIS: ftp://cddis.gsfc.nasa.gov/pub/gps/products/mgex/dcb/

*BIAS	SVN	PRN	STATION	OBS1	OBS2	BIAS_START	BIAS_END	UNIT	ESTIMATED_VALUE	STD_DEV
DSB	G063	G01		C1C	C1W	2022:263:00000	2022:264:00000	ns	-1.0030	0.0060
DSB	G061	G02		C1C	C1W	2022:263:00000	2022:264:00000	ns	1.7610	0.0060
DSB	G069	G03		C1C	C1W	2022:263:00000	2022:264:00000	ns	-1.3640	0.0060
DSB	G074	G04		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.7660	0.0060
DSB	G050	G05		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.7660	0.0060
DSB	G067	G06		C1C	C1W	2022:263:00000	2022:264:00000	ns	-1.2870	0.0060
DSB	G048	G07		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.3280	0.0060
DSB	G072	G08		C1C	C1W	2022:263:00000	2022:264:00000	ns	0.2490	0.0060
DSB	G068	G09		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.0780	0.0060
DSB	G073	G10		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.2870	0.0060
DSB	G078	G11		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.9450	0.0060
DSB	G058	G12		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.2560	0.0060
DSB	G043	G13		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.1150	0.0060
DSB	G077	G14		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.8480	0.0060
DSB	G055	G15		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.8320	0.0060
DSB	G056	G16		C1C	C1W	2022:263:00000	2022:264:00000	ns	1.1870	0.0060
DSB	G053	G17		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.3370	0.0060
DSB	G075	G18		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.9280	0.0060
DSB	G059	G19		C1C	C1W	2022:263:00000	2022:264:00000	ns	2.4550	0.0060
DSB	G051	G20		C1C	C1W	2022:263:00000	2022:264:00000	ns	2.0800	0.0060
DSB	G045	G21		C1C	C1W	2022:263:00000	2022:264:00000	ns	2.0230	0.0060
DSB	G041	G22		C1C	C1W	2022:263:00000	2022:264:00000	ns	0.5940	0.0060
DSB	G076	G23		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.8720	0.0060
DSB	G065	G24		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.7690	0.0060
DSB	G062	G25		C1C	C1W	2022:263:00000	2022:264:00000	ns	0.9360	0.0060
DSB	G071	G26		C1C	C1W	2022:263:00000	2022:264:00000	ns	0.1740	0.0060
DSB	G066	G27		C1C	C1W	2022:263:00000	2022:264:00000	ns	0.3180	0.0060
DSB	G044	G28		C1C	C1W	2022:263:00000	2022:264:00000	ns	1.1720	0.0060
DSB	G057	G29		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.3280	0.0060
DSB	G064	G30		C1C	C1W	2022:263:00000	2022:264:00000	ns	0.8230	0.0060
DSB	G052	G31		C1C	C1W	2022:263:00000	2022:264:00000	ns	-0.6070	0.0060
DSB	G070	G32		C1C	C1W	2022:263:00000	2022:264:00000	ns	-1.0560	0.0060
DSB	G063	G01		C2W	C2S	2022:263:00000	2022:264:00000	ns	1.3970	0.0085



Smead Aerospace

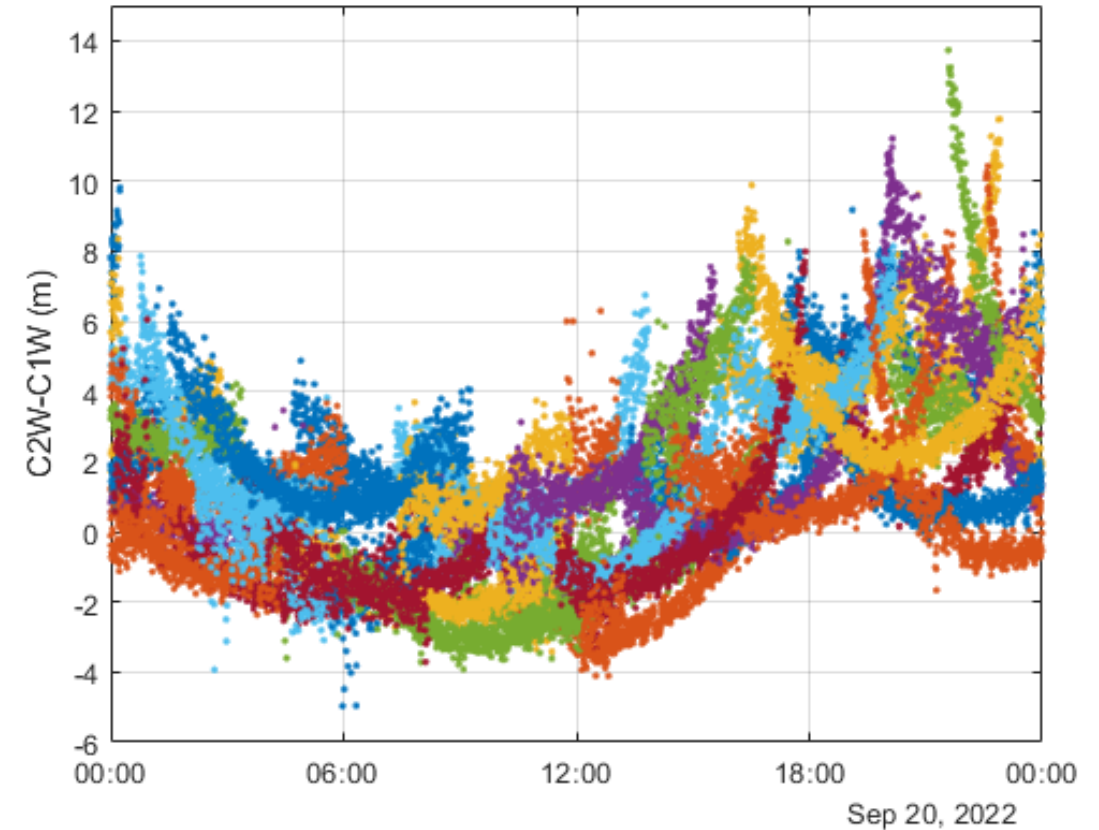
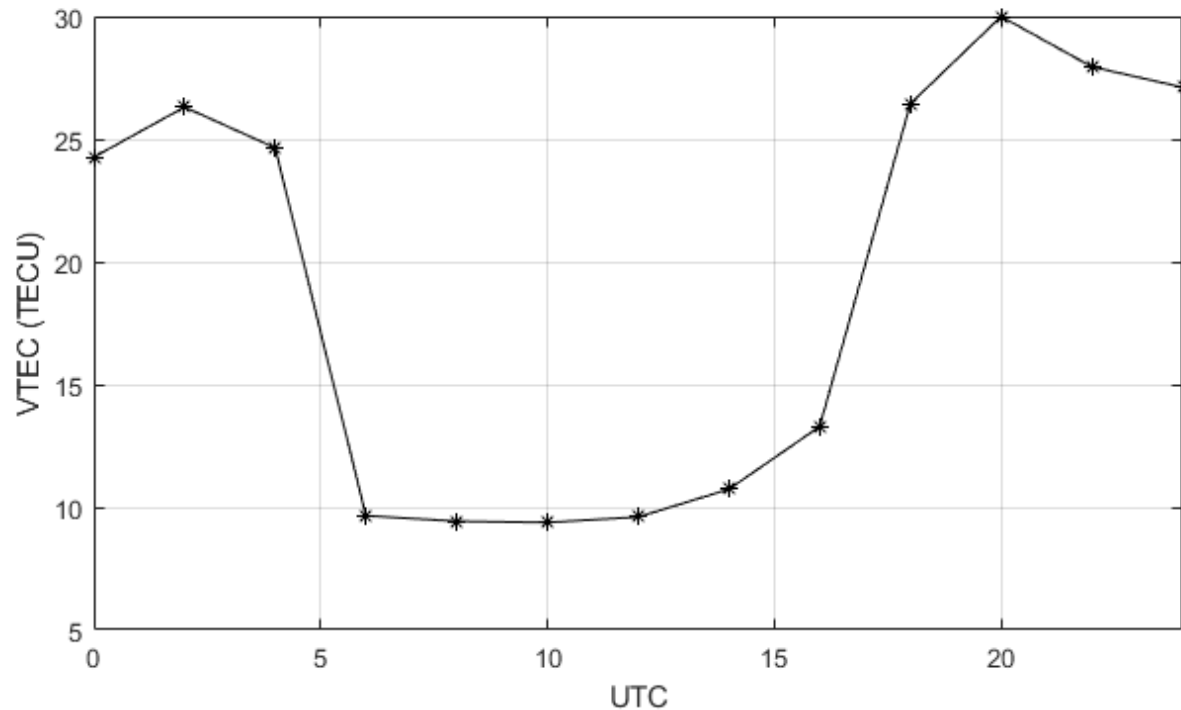
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https://cddis.nasa.gov/Data_and_Derived_Products/GNSS/gnss_differential_code_bias_product.html

ASEN5090 Fall 2025

Receiver DCB

- Interpolated VTEC using the IGS GIM for NIST
- Again, 1 TECU corresponds to 0.16 cm L1 signal group delay



- Receiver DCB impacts the ionospheric delay/TEC estimation but does not impact dual-freq delay positioning
- It does cause errors in time transfer

TROPOSPHERE DELAY MODEL – UNB3m

<https://gge.ext.unb.ca/Resources/unb3m/unb3m.html>

Table 1. Look-up table of UNB3 model.

Average					
Latitude (degrees)	Pressure (mbar)	Temperature (K)	WVP* (mbar)	β (K m ⁻¹)	λ (-)
15	1013.25	299.65	26.31	6.30e-3	2.77
30	1017.25	294.15	21.79	6.05e-3	3.15
45	1015.75	283.15	11.66	5.58e-3	2.57
60	1011.75	272.15	6.78	5.39e-3	1.81
75	1013.00	263.65	4.11	4.53e-3	1.55
Amplitude					
Latitude (degrees)	Pressure (mbar)	Temperature (K)	WVP* (mbar)	β (K m ⁻¹)	λ (-)
15	0.00	0.00	0.00	0.00	0.00
30	-3.75	7.00	8.85	0.25e-3	0.33
45	-2.25	11.00	7.24	0.32e-3	0.46
60	-1.75	15.00	5.36	0.81e-3	0.74
75	-0.50	14.50	3.39	0.62e-3	0.30

$$X_{\phi, \text{doy}} = \text{Avg}_{\phi} - \text{Amp}_{\phi} \cdot \cos\left((\text{doy} - 28) \frac{2\pi}{365.25}\right)$$

$$d_h^z = \frac{10^{-6} k_1 R}{g_m} \cdot P_0 \cdot \left(1 - \frac{\beta H}{T_0}\right)^{\frac{g}{R\beta}}$$

$$d_{nh}^z = \frac{10^{-6} (T_m k'_2 + k_3) R}{g_m \lambda' - \beta R} \cdot \frac{e_0}{T_0} \cdot \left(1 - \frac{\beta H}{T_0}\right)^{\frac{\lambda' g}{R\beta} - 1}$$

(Leandro et al., 2006)

UNB3 uses mapping functions developed by A. E. Niell (1996)

UNB3m_pack matlab package available

[[RTROP HZD HMF WZD WMF]= UNB3M(LATRAD,HEIGHTM,DAYOYEAR,ELEVRAD)



Saastomoinen model

$$\tilde{T}_{z,d} = 0.002277(1 + 0.0026 \cos(2\varphi) + 0.00028H)P_0 \quad (\text{mm})$$

$$\tilde{T}_{z,w} = 0.002277 \left(\frac{1255}{T_0} + 0.05 \right) e_0 \quad (\text{mm})$$

Values at RX antenna
based on local measurements
or Standard Atmosphere model

f : latitude
 H : Antenna height above sea level (km) (orthometric height)
 P_0 : total pressure (mB)
 e_0 : partial water vapor pressure (mB)
 T_0 : temperature (K)

$$e_0 = 6.108 \times RH \times e^{\frac{17.15T_0 - 4684}{T_0 - 38.45}}$$

↑ Relative humidity (it is a percentage value)

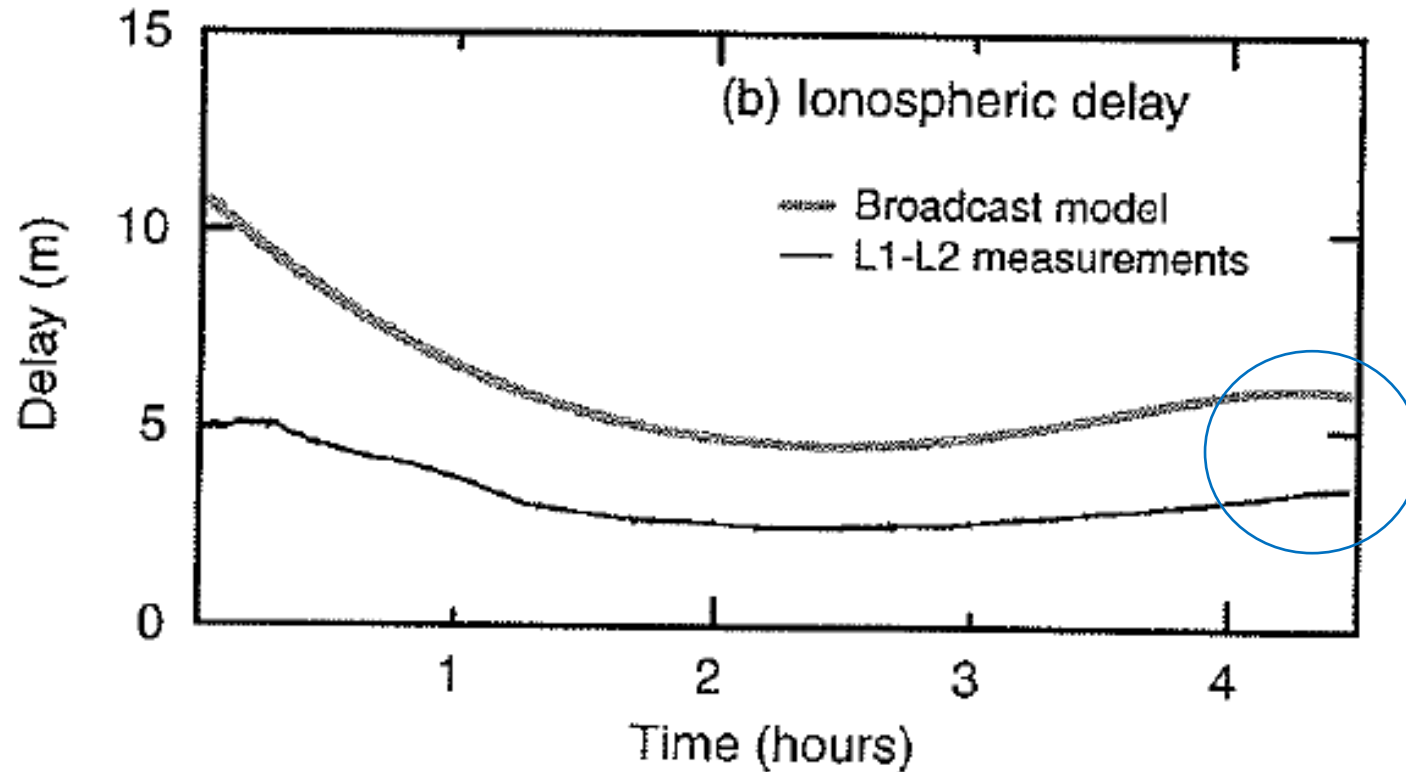
TROPOSPHERE DELAY MODEL – Other Info

- In-situ weather measurements
 - P0: total pressure (mB)
 - e0: partial water vapor pressure (mB) or relative humidity
 - T0: temperature (K)
- Regional GNSS network measurement of tropo. delay
 - Similar idea with differential/SBAS
- Modeling
 - UNB3 (coarse but convenient)
 - GPT3/VMF3 (more comprehensive model, good mapping function, 6-hour 1-degree ZHD & ZWD available)
 - ECMWF Reanalysis V5 model and forecast (hourly, 0.25-degree resolution)



Empirical Data: Ionosphere Delay

L1-L2 measurements are the differential carrier phases with integer ambiguity removed using code phase measurements. The data was taken on a quiet day.



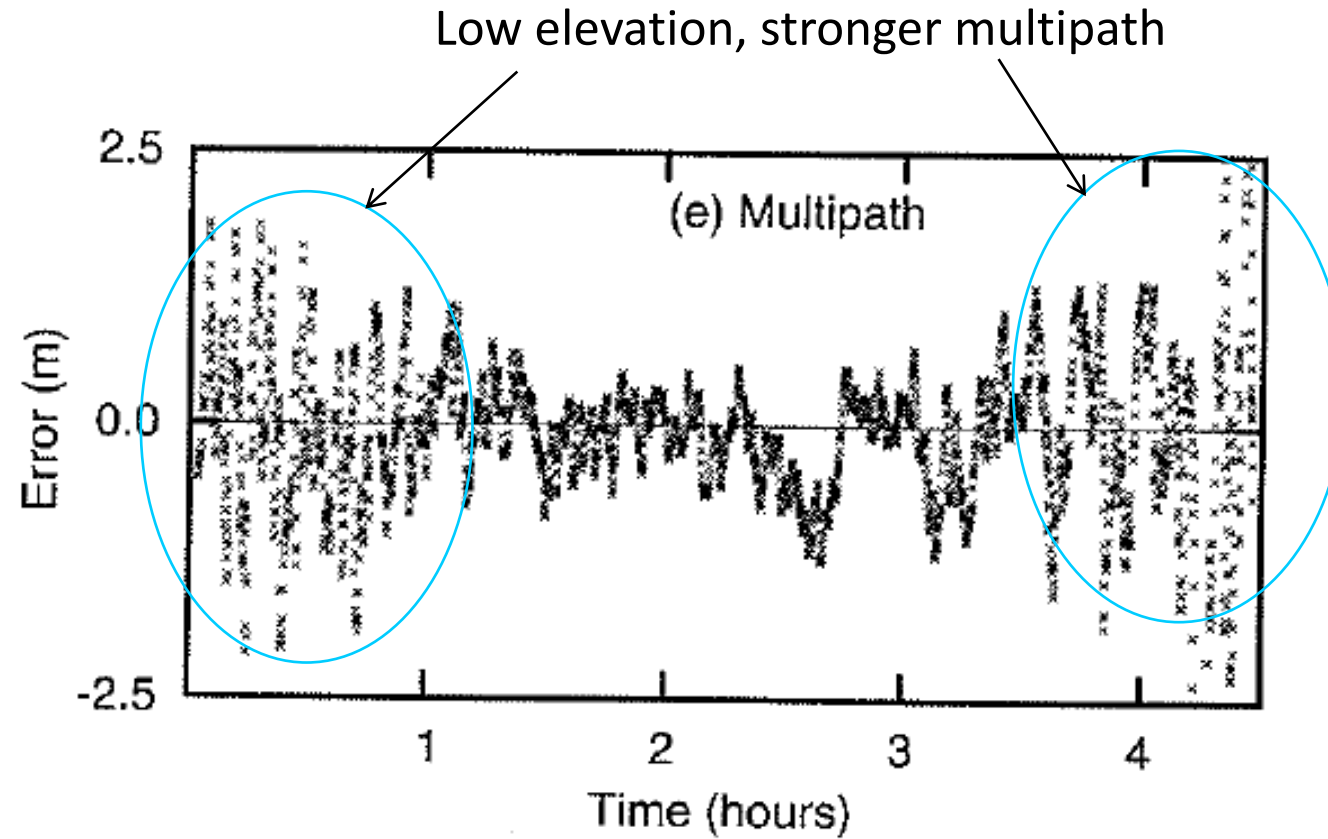
SV was setting
on the north-east
away from the sun

Pseudorange Modeling to Correction

- Pseudorange modeling
 - uses a known or an expected receiver location
 - Isolates each contributor to the pseudorange error and analyzes the ranging accuracy
- The observed pseudorange corrected for all the errors you know:
 - satellite clock model
 - relativistic model
 - ionospheric model, grid lookup, or dual frequency correction
 - tropo model
- What errors are left?
 - Noise
 - Multipath
 - etc.



Empirical Data: Multipath and RX Noise



Code multipath observable

- To evaluate multipath on GPS observables, you need a way to remove all other error sources (orbits, ionosphere, relativity troposphere, clocks).

- With dual frequency observations compute MP1 or MP2:

$$MP_1 = \rho_1 - \frac{f_1^2 + f_2^2}{f_1^2 - f_2^2} \phi_1 \lambda_1 + \frac{2f_2^2}{f_1^2 - f_2^2} \phi_2 \lambda_2 \approx \tau_{M_1} + C_0 + \epsilon_{\rho_1}$$

- With only single frequency observations Code minus carrier (CMC) is used to approximate the pseudorange multipath (ignoring contributions of carrier multipath and noise)

$$CMC_1 = \rho_1 - \phi_1 \lambda_1 \approx \tau_{M_1} + 2I_1 + C_0 + \epsilon_{\rho_1}$$

Code tracking error due to noise in a coherent DLL

$$\sigma_{\Delta\tau} = T_c \sqrt{\frac{d}{4T C / N_0}} = T_c \sqrt{\frac{B_{\tau,1} d}{2 C / N_0}}$$

where

d is the spacing between the early and late correlators

T is the coherent integration time

T_c is the duration of a chip (one over the chipping rate)

$B_{\tau,1}$ is the noise equivalent bandwidth of the DLL tracking loop

Typical numbers:

$T_c \sim 300\text{m}$, $d = 1$, $T = 1\text{s}$,

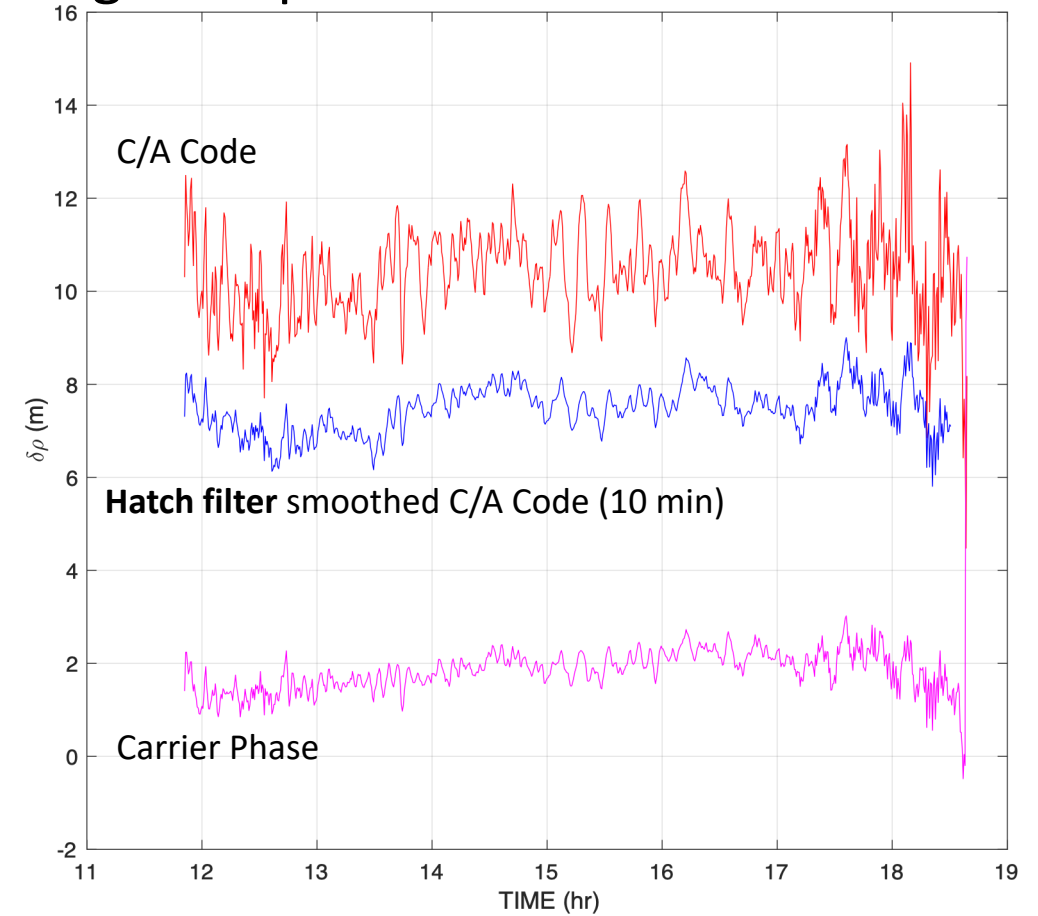
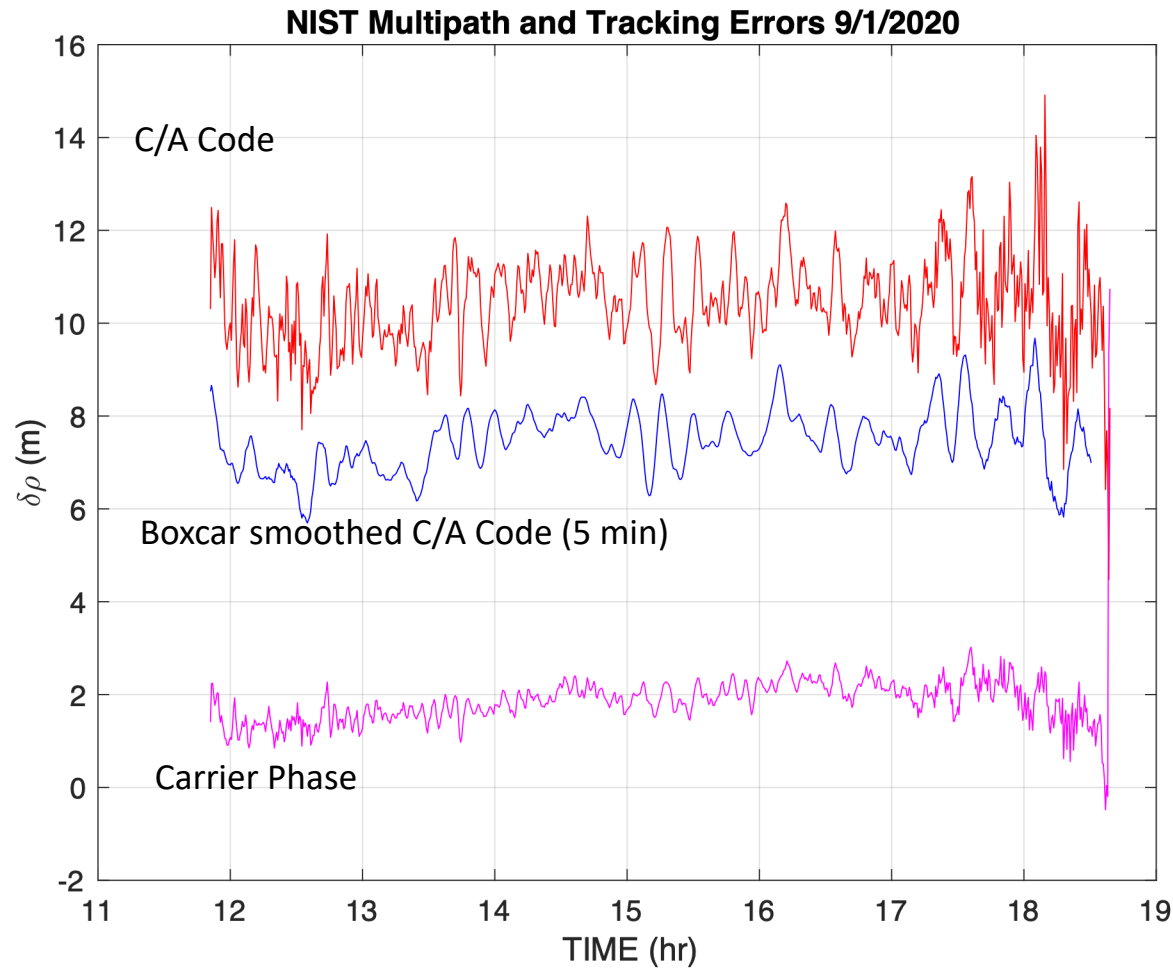
$C/N_0 = 1\text{e}4 \text{ Hz}$ (weird units!)

$\sigma = 300\text{m} * 5\text{e-}3 = 1.5\text{m}$ *approximately*

Higher tracking loop bandwidth produces noisier measurements, but is more responsive to signal dynamics due to vehicle motion, scintillation, or clock noise.

Carrier smoothing to reduce code multipath & noise

Ignoring ionospheric effects for now....



$$\bar{\rho}_i = \frac{1}{M} \rho_i + \frac{M-1}{M} (\bar{\rho}_{i-1} + \phi_i - \phi_{i-1})$$

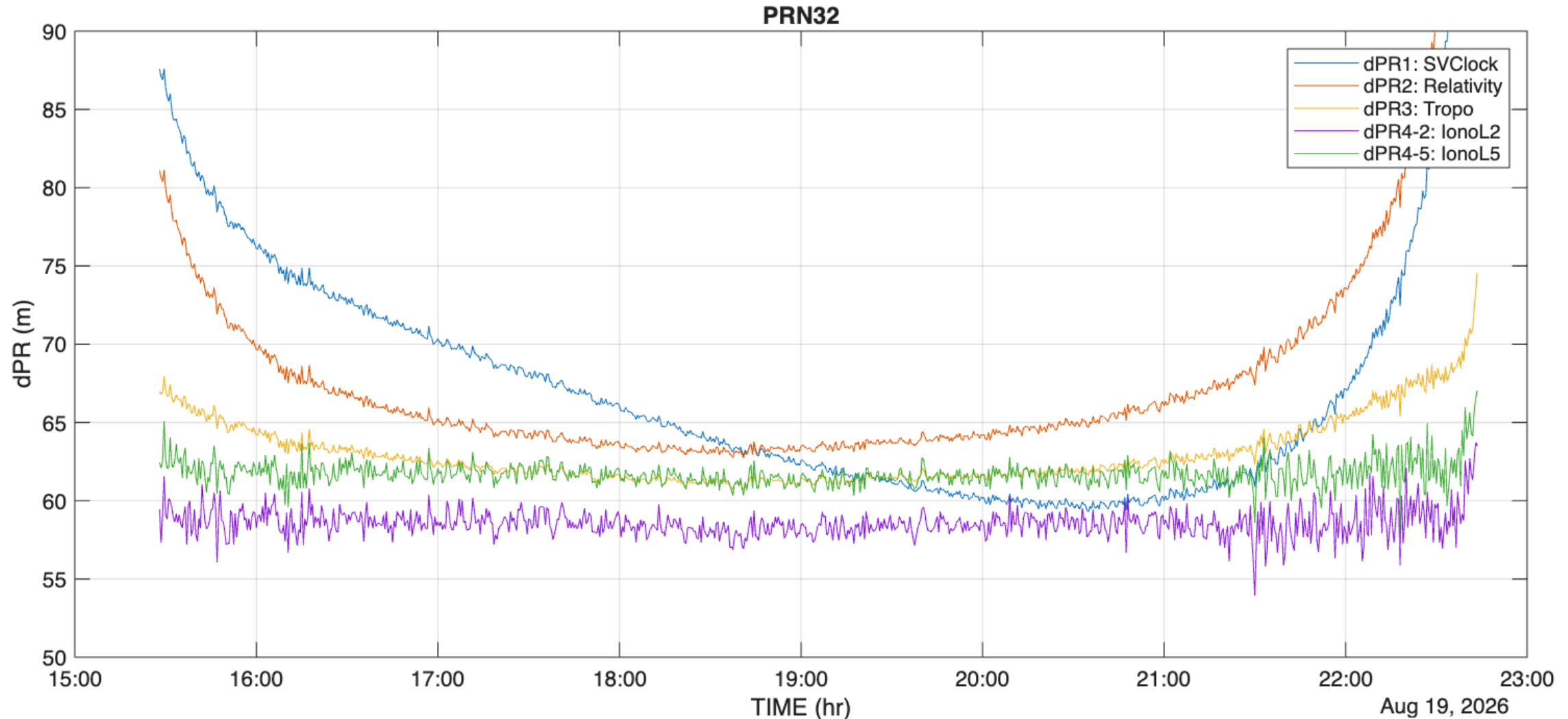


Measurement Residual

- Difference between actual measurement & expected/predicted measurement
- Actual measurement is based on observation(s) reported by the receiver
- Expected measurement is based on:
 - Reported satellite position
 - Assumed (or a priori) user position (& possibly user clock)
 - Delays that can be modeled: satellite clocks, tropo, iono
- Measurement residuals used to solve for corrections to assumed user position & clock.
- Solution is performed using a least-squares approach.

HW4 – Sequentially applying pseudorange Corrections

Example of pseudorange residuals for PRN22 observed at NIST for a data set in 2026



Differential GPS (DGPS)

- Pseudorange errors that are common to receivers in the same area can be reduced using DGPS
- A reference receiver at a known location can transmit range corrections (via separate comm link) to nearby users.
- The amount of correction depends on how quickly the errors decorrelate in space and time
- In the days of Selective Availability this was a very big deal
- With single frequency receivers this is also important
- Now, clock and orbit errors can be largely corrected using precise data products and dual frequency receivers are becoming more common.



Differential GPS

- Pseudorange observables

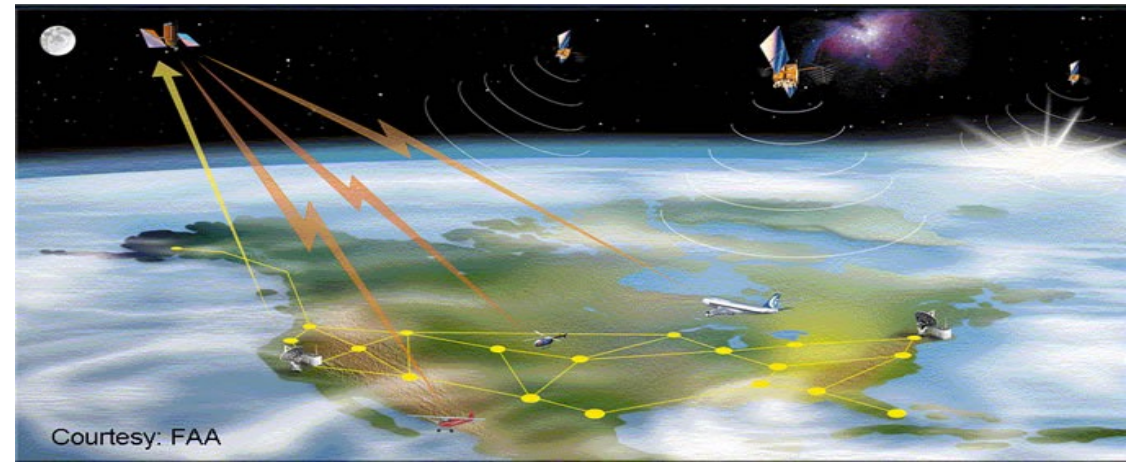
$$\rho_R = R_R + c(\delta t_R - \delta t^s) + T_R + I_R + M_R + \varepsilon_R$$

$$\rho_U = R_U + c(\delta t_U - \delta t^s) + T_U + I_U + M_U + \varepsilon_U$$

- At reference station, compute pseudorange error $\delta\rho_R = \rho_R - \hat{R}_R = c(\delta t_R - \delta t^s) + O_R + T_R + I_R + M_R + \varepsilon_R$ and send to local users:
- In the user receiver, apply to measurements: $\rho_{Ucorrected} = \rho_U - \delta\rho_R$
$$\rho_{Ucorrected} = R_U + c(\delta t_U - \delta t_R) + (O_U - O_R) + (I_U - I_R) + (T_U - T_R) + (M_U - M_R) + (\varepsilon_U - \varepsilon_R)$$
- Orbit (ephemeris) errors, iono, and tropo experienced by reference and user receivers are similar/correlated.
 - Iono decorrelation is $\sim 1\text{m}/100\text{km}$, tropo depends on height & weather
 - Smaller separation and more frequent corrections yield better accuracy
- Multipath and tracking errors are not correlated between stations

WAAS/SBAS

- Developed for civil aviation by the [Federal Aviation Administration](#)
- Provides integrity & 3 m accuracy 95% of the time
- 25 ground stations throughout the US and Alaska
- Measurements sent to master station that estimates correction vector
- Corrections are uploaded to two geostationary satellites
- WAAS-enabled GPS receivers interpolate and apply the corrections to their measured ranges.
- [European Geostationary Navigation Overlay Service](#) (EGNOS)
- Japanese [Multi-Functional Satellite Augmentation System](#) (MSAS)



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Next lecture: Least Squares Solutions